

Usman Farooq

✉ Usmanfarooq5445@gmail.com

☎ 03250259200

📍 Punjab University town 2, Lahore.

🌐 <https://www.linkedin.com/in/usman-farooq-j0259200>



Profile

Metallurgy and Materials Engineering graduate with hands-on experience in wastewater flow monitoring, non-destructive testing (UT, DPT, MT), and industrial quality assurance. Served as an On-Site Engineer at Welcos Pvt Ltd until February 2026, contributing to large-scale wastewater survey projects at Sunder Industrial Estate. Skilled in SEM, XRD, FTIR, electrochemical testing, and metallography. Strong background in materials characterization, process optimization, and industrial safety practices.

Professional Experience

➤ On-Site Engineer – Welcos Private Limited.

01/10/2025 – 02/2026

- ❑ Executing wastewater flow monitoring surveys at Sunder Industrial Estate covering multiple industrial units.
- ❑ Installed and calibrated flow measurement devices ensuring accurate real-time data acquisition.
- ❑ Maintained 98%+ data accuracy through systematic recording and validation procedures.
- ❑ Coordinated with cross-functional teams to ensure timely project execution and compliance with safety standards.

➤ Internee – SGS Pakistan Private Limited

02/07/2024 – 30/7/2024

- Assisted in performing Ultrasonic Testing (UT), Dye Penetrant Testing (DPT), and Magnetic Particle Testing (MT) under certified inspectors.
- Supported inspection of industrial components to ensure compliance with safety and quality standards.
- Observed cathodic protection system evaluation for corrosion prevention.

Education

BSc Metallurgy and Materials Engineering,

University of the Punjab, Lahore

Intermediate Pre Engineering,

Pak Garrison College, Nankana Sahib

Matriculation Science Group,

Govt.High School, Nankana Sahib

Certifications & Workshops

- **ISO 14001:2015 Environmental Management System Awareness** — *The Knight of Safety*
- **The COSHH Risk Assessor Certification** — *The Knight of Safety*
- Globally Responsible Engineering Job Simulation – Engineers Without Borders UK (Forage).
- **Explore Supply Chain Job Simulation** – Forage
- **Workshop:** Computational Materials Engineering
- **Materials Revolution:** Advance Materials for Sustainable Future.
- **ISAM-2025** 19th International Symposium on Advance Materials
- **10th** Invention to Innovation Summit 2025.
- **9th IFCE2025** – International Foundry Congress & Exhibition

Technical Skills

- **Industrial & Lab Skills:** SEM, XRD, FTIR, EDX, AFM, Electrochemical Testing, Mechanical Testing (tensile, hardness, compression, impact, creep, fatigue).
- **Engineering Skills:** Metallography, Heat Treatment, CAD, Engineering Drawings, Mineral Processing, Foundry Engineering.
- **Process & Quality Skills:** R&D, Data Analysis, Quality Control, Industrial Exposure.
- **Software:** Microsoft Office, Origin, AutoCAD Tools.

Projects

- **Synthesis of Zeolite for N₂/O₂ Separation (Final Year Project)** – Synthesized and tested zeolite adsorbents for nitrogen/oxygen separation applications.
 - **Hydrogen Embrittlement in Grey Cast Iron** – Investigated embrittlement effects in automotive components.
 - **Steel Casting Defects with Sand Additives** – Analyzed effect of additives on defect reduction in steel castings.
 - **Bentonite Clay & Moisture Effects** – Studied influence of clay and moisture on molding sand strength.
 - **Impurity Elements in Semiconductors** – Researched impact of impurities on charge transport in semiconductor materials.
- Metallography of Spring Steel Wire Samples** – Compared master alloy and scrap-derived SAE 1062B spring steel; identified differences in microstructure and carbide distribution affecting mechanical performance.
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Core Competencies

- Process Optimization & Industrial Operations
 - Mechanical & Materials Testing
 - Quality Assurance & Control
 - Problem Solving & Logical Thinking
 - Team Collaboration & Communication
 - Organizational & Planning Skills
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Languages

- English
- Urdu
- Punjabi